



Current Ramping: A Strategy for Rapid Start-up of PEMFCs from Subfreezing Environment

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A rapid start-up of proton exchange membrane fuel cells (PEMFCs) from subzero temperatures is essential for fuel cell vehicle commercialization. We explore the limits of the start-up time and the required cell material properties and operating conditions using an experimentally validated model. A linear class of current-ramping protocols is proposed and optimized for a rapid cold start. A PEMFC with a standard cell thermal mass of 0.4 J/cm² K starting from −30°C is of primary interest and is extensively studied in this work. Either a small initial current density (e.g., 100 mA/cm²) combined with an intermediate ramping rate or a relatively large initial current density (e.g., 200 mA/cm²) in combination with a small ramping rate can lead to a successful self-start if the membrane electrode assembly (MEA) is sufficiently dry before the start-up. However, a current-ramping cold start using an insufficient initial current density (e.g., ≤50 mA/cm²) shuts down with any ramping rate. A more rapid self-start can be achieved by increasing the initial current density, which is mainly limited by the initial water content in the MEA. Hence, keeping the MEA mildly hydrated before the cold start can be favorable to a rapid start-up of a PEMFC using current ramping. This strategy is particularly effective for the rapid start-up of next-generation PEMFCs with reduced thermal mass. A PEMFC with a thermal mass of 0.2 J/cm² K and a relatively wet MEA can be successfully started from −30°C in about 5 s by applying an initial current density of as high as 1 A/cm².

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Fundamental mechanisms for a cold start of a proton exchange membrane fuel cell (PEMFC) have been clarified by a recent series of experimental and theoretical studies.^{1–17} The theory of intraelectrode ice formation (IIF) has been put forth to explain experimental observations and forecast cold start performance over a wide range of membrane electrode assembly (MEA) properties and cell operating conditions.¹⁸ However, a rapid self-start of a PEMFC engine from a subfreezing environment, particularly from temperatures around −30°C, remains a challenge for automotive applications. According to the IIF theory, the fate of a subzero start-up is ultimately determined by the competition between heat generation and ice formation in the cathode catalyst layer (CCL). An optimal rapid self-start thus requires that the heat generation within the cell be maximized and that the cell be warmed up quickly to the freezing point while simultaneously keeping ice formation in the CCL below the shutdown threshold.

Ice formation in the CCL during a cold start is a net outcome of water flow inside the cell. Figure 1 schematically shows water production and transport on the cathode side of a PEMFC, where the oxygen reduction reaction (ORR) produces water in the CCL; water on the anode side may be dragged to the CCL due to electro-osmotic drag (EOD). Water removal from the CCL mainly relies on two mechanisms: (i) vapor-phase transport bringing some water vapor into the cathode gas diffusion layer (CGDL) and gas channel with some water emitting to the ambient along with the exhaust gas and (ii) water diffusing into the membrane via back-diffusion. The ionomers in the CCL can uptake some water as well if they are initially dry. Excess water accumulating in the CCL and CGDL desublimates into ice/frost once the local water vapor becomes oversaturated. As the vapor saturation pressure over ice is very low at a subzero temperature, little ice is expected to form in the CGDL.

The total heat generation rate, $\dot{Q}_{\text{total}}$, during a start-up process is roughly proportional to the current density such that

$$\dot{Q}_{\text{total}} \approx (E_h - U_{\text{cell}})I \quad [1]$$

where E_h denotes the thermodynamic potential of the H₂–O₂ reaction, with the product water being ice. In a current-ramping start-up, the current density is typically low in the initial stages, which offers

time for absorption of product water into the initially dry membrane. Then, the increasing current density over time accelerates the heat generation for cell warm-up, possibly yielding a rapid self-start.

Two key parameters describing a linear class of current-ramping cold start are the initial current density and the current-ramping rate. Using a numerical model previously validated extensively by experimental data, a series of parametric studies was performed in this work to explore the optimum parameter set that can result in rapid current-ramping cold starts. In the following section, the numerical model is briefly introduced. Results from extensive parametric studies are then presented and discussed in detail. Lastly, major conclusions are summarized.

Numerical Model

The present work is a further application of the model reported in Ref. 13. This nonisothermal, multiphase model fully accounts for the transient transport and electrochemical processes with ice formation and can be concisely summarized in the form of the following governing equations

$$\frac{\partial (\varepsilon_s \rho_s)}{\partial t} + \frac{\partial (\varepsilon \rho)}{\partial t} + \nabla (\rho \mathbf{u}) = 0 \quad [2]$$

momentum conservation

$$\frac{\partial (\rho \mathbf{u} / \varepsilon)}{\partial t} + \nabla \left(\frac{\rho \mathbf{u} \mathbf{u}}{\varepsilon^2} \right) = \nabla (\mu \nabla \mathbf{u}) - \nabla P + S_u \quad [3]$$

species conservation

$$\frac{\partial (\varepsilon C^i)}{\partial t} + \nabla (\mathbf{u} C^i) = \nabla (D_{\text{eff}}^i \nabla C^i) + S_C^i \quad [4]$$

charge conservation (electrons)

$$0 = \nabla (\sigma_s^{\text{eff}} \nabla \phi_s) + S_{\phi_s} \quad [5]$$

and charge conservation (protons)

$$0 = \nabla (\kappa_e^{\text{eff}} \nabla \phi_e) + S_{\phi_e} \quad [6]$$

In Eq. 2, ε_s and ρ_s denote ice fraction and density of ice, respectively, which differ from ε and ρ , denoting porosity and density of fluid, respectively. The subscripts “s” used in Eq. 5 indicate the solid phase, which is different from the usage of subscript “s” in Eq. 2.

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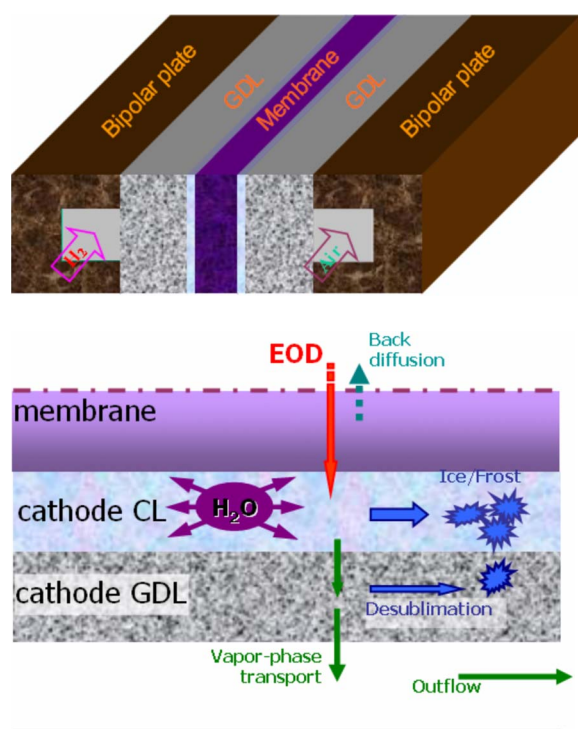


Figure 1. (Color online) Schematic illustration of a single-cell PEMFC and the involved water flow during cold start.

For more details about the nomenclature, one may refer to Ref. 13.

Lumped Thermal Model

We assume that the cell has been prepared with an equilibrium purge before its start-up,⁶ resulting in uniform initial membrane water content. Such a situation justifies the existence of a uniform temperature throughout the cell and hence the use of a lumped thermal model.

In the lumped thermal model, the total heat generation rate during cold start is simply the volumetric integration of all local heat sources, such that

$$\dot{Q}_{\text{total}} = \frac{1}{A} \int_{V_{\text{catalyst layers}}} \left[j \left(\eta + T \frac{dU_0}{dT} \right) + \frac{i_e^2}{\kappa_e^{\text{eff}}} + \frac{i_s^2}{\sigma_s^{\text{eff}}} + \dot{q}_{\text{gs}}^{\text{H}_2\text{O}} h_{\text{gs}} \right] dV + \frac{1}{A} \int_{V_{\text{mem}}} \frac{i_e^2}{\kappa_e} dV \quad [7]$$

Heat losses from the cell to the surroundings include two contributions: (i) heat loss due to exhaust gases and (ii) convective heat loss from outer surfaces of the cell's bipolar plates. Mathematically, this can be expressed as

$$\dot{Q}_{\text{loss}} = \frac{1}{A} [(\dot{m}C_p)_{\text{out}}T - (\dot{m}C_p)_{\text{in}}T_0] + h(T - T_0) \quad [8]$$

where h is the heat convection coefficient on the cell outer surfaces in $\text{W/m}^2 \text{ K}$. Applying energy balance to the cell and invoking the assumption of a spatially uniform temperature, the cell temperature can be calculated from

$$(\dot{m}C_p)_{\text{cell}}(T - T_0) = \int_0^t (\dot{Q}_{\text{total}} - \dot{Q}_{\text{loss}}) dt \quad [9]$$

where the cell thermal mass, $(\dot{m}C_p)_{\text{cell}}$, has a unit of $\text{J/cm}^2 \text{ K}$, which is the volumetric sum of the heat capacities of individual compo-

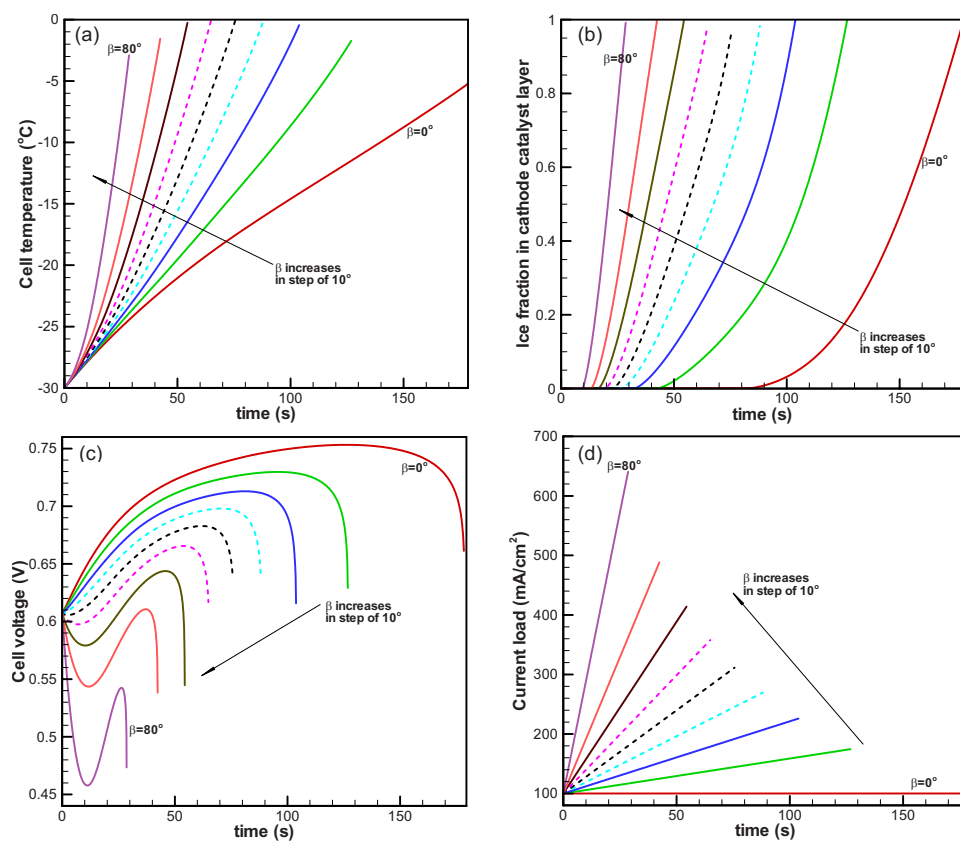


Figure 2. (Color online) Current-ramping start-ups with initial current density of 100 mA/cm^2 . (a) Cell temperature vs time, (b) ice fraction in CCL vs time, (c) cell voltage vs time, and (d) current density vs time. Note: Dashed lines indicate successful start-up cases, and solid lines indicate failed start-up cases.

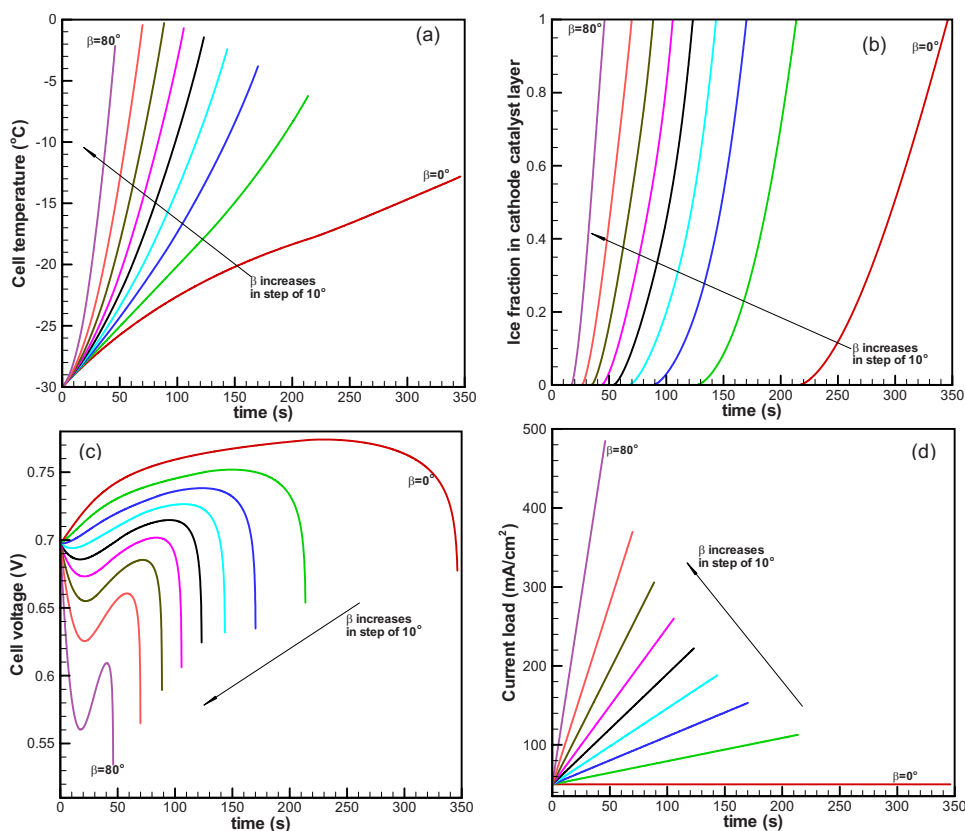


Figure 3. (Color online) Current-ramping start-ups with initial current density of 50 mA/cm². (a) Cell temperature vs. time, (b) ice fraction in CCL vs. time, (c) cell voltage vs. time, and (d) current density vs. time. Note: All the start-up cases failed.

nents divided by the electrode active area. The spatially uniform cell temperature given by Eq. 9 is fully coupled to other conservation equations with the assumption of instantaneous ice formation.

The numerical details of solving Eq. 2-6 in conjunction with Eq. 9 are found in Ref. 7 and 13. These nonisothermal, multiphase, and transient simulations can be run routinely on a single PC, as evident from extensive parametric studies presented below.

Results and Discussion

The goals of the present work are (i) to seek an optimum parameter set that enable a rapid current-ramping start-up and (ii) to provide a practical design of optimal start-up strategies for fuel cell vehicle development. Therefore, detailed spatial distributions of ice fraction, current density, and membrane water content are not the focus of this work. Rather, we focus on elucidating the dynamic evolutions of cell temperature and ice fraction at the center of the CCL and on quantitatively evaluating the effects of current ramping on PEMFC start-up. We define the requirements for a successful current-ramping cold start as follows: The cell temperature is elevated to 0°C, while the ice fraction at the center of the CCL remains below unity.

In the present work, the current density during cold start is ramped up according to the following dimensionless form

$$\frac{I}{I_0} = 1.0 + \tan(\beta) \frac{t}{30} \quad [10]$$

where I and I_0 are the current density at time t (in seconds) and the initial instant, respectively, and β is the current-ramping angle, and hence $\tan(\beta)$ represents the ramping rate. The 30 s critical time used to nondimensionalize t is taken from a technical target identified by the U.S. Department of Energy: In year 2010, PEMFC stacks must start up from -20°C to a maximum power in 30 s.¹⁹

In the present work, starting a PEMFC having a cell thermal mass (mC_p) of 0.4 J/cm² K from the -30°C environment is taken to be the baseline. Geometrical parameters and electrochemical,

physical, and transport properties remain the same as those in Ref. 13 and thus are not repeated here. The convective heat coefficient (h) is assumed to be 10.0 W/m² K to mimic an end cell in a stack but obviously can be varied to represent middle cells as well (e.g., $h = 0$ would represent an adiabatic condition). Before cold start, we assume that a sufficiently long purge⁶ has been performed such that the MEA is free of ice, and the membrane is dry with an initial water content λ_0 equal to 3.0.

$I_0 = 100$ mA/cm².— The results of a baseline set of simulations with the current-ramping angle β varying from 0 to 80° are shown in Fig. 2. As defined in Eq. 10, the larger the angle β , the higher the ramping rate. In Fig. 2, the dashed lines represent a successful self-start as marked by the cell temperature reaching the freezing point, while the ice fraction in the CCL remains below unity. A successful start-up is realized only for the intermediate values of β : 30–50°. If β is smaller than 20°, an overly slow ramping current density cannot yield a sufficiently rapid cell temperature rise, thus leading to a shutdown after the CCL becomes full of ice. However, if β is larger than 60°, ramping current density too quickly results in water production being too fast, with insufficient time for water absorption into the membrane. This causes a shutdown once the CCL is full of ice before the cell temperature rises to 0°C. The parametric study presented here clearly illustrates the fact that the competition between ice formation and heat generation determines the fate of a cold start process.

Water accumulation in the CCL stems from two sources. One is the water production from ORR, and the other is EOD drag from the anode catalyst layer. Theoretically, the water production rate and EOD drag are both linearly dependent on the current density I as follows

$$\dot{n}_{\text{pro}} = \frac{M^{\text{H}_2\text{O}} I}{2F} \quad \dot{n}_{\text{EOD}} = - \frac{M^{\text{H}_2\text{O}} n_d I}{F} \quad [11]$$

where n_d is a constant EOD coefficient chosen to be at unity in the present work, which is confirmed by the most recent experimental

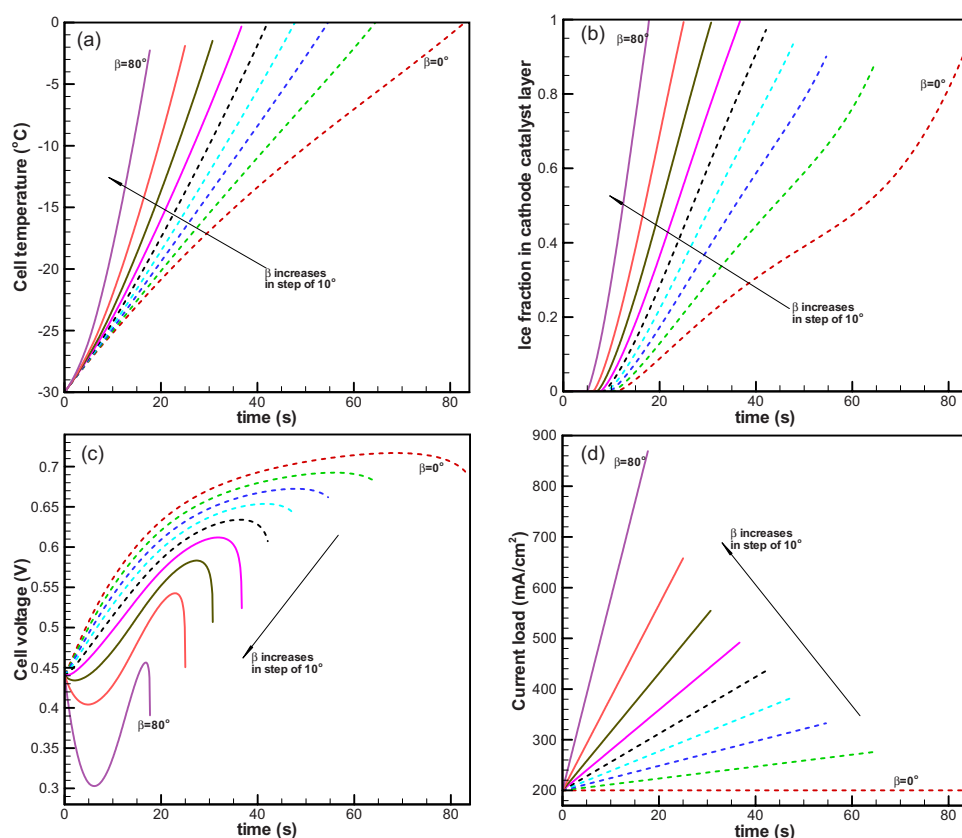


Figure 4. (Color online) Current-ramping start-ups with initial current density of 200 mA/cm². (a) Cell temperature vs time, (b) ice fraction in CCL vs time, (c) cell voltage vs time, and (d) current density vs time. Note: Dashed lines indicate successful start-up cases, and solid lines failed start-up cases.

data.²⁰ However, water absorption into the membrane mainly relies on the water diffusivity and water uptake of the membrane. Membrane water diffusivity is a strong function of temperature, as reported by Jiang et al.¹³

$$D_{\text{mem}}^{\text{H}_2\text{O}} = \begin{cases} 5.93 \times 10^{-5} \lambda (e^{0.28\lambda} - 1) e^{(-4269/T)} & \text{for } 0 < \lambda \leq 3 \\ 7.97 \times 10^{-6} \lambda (1 + 161e^{-\lambda}) e^{(-4269/T)} & \text{otherwise} \end{cases} \quad [12]$$

The above equation implies that the ability of water to be absorbed into the membrane improves with the cell temperature. Therefore, one can ramp up the current density and hence water sources via ORR and EOD without significantly increasing ice accumulation in the CCL. Thus, the essence of current-ramping start-up is to gradually ramp up the applied current density with cell temperature. This is the reason that an optimal range of ramping rates exists, as evident from Fig. 2. In the present work, we extended the membrane water uptake curve measured by Springer et al.²¹ at 30°C to the subzero temperature range as a first approximation due to the absence of relevant data. After the present work was completed, we noticed the publication of Gallagher et al.²⁰ in which membrane water uptake was measured at subzero temperatures and the maximum water content at full hydration decreases with decreasing temperatures. This implies that the water storage capacity of the membrane may be smaller than that predicted using Springer's water uptake curve,²¹ and cold start becomes more difficult than predicted by the present simulations. Nonetheless, qualitative trends of the simulations and fundamentals of the current-ramping method presented in this work remain valid.

The cell voltage curves plotted in Fig. 2c illustrate to some extent the competition between heat generation and ice formation during cold start. A major voltage loss during cold start is from the ohmic transport due partly to the exceedingly low temperatures and partly to the relatively dry membranes needed for self-start. The ohmic loss is directly proportional to the current density I and inversely to the

membrane proton conductivity κ_e . The latter is a strong function of both membrane water content and cell temperature, as shown by Jiang et al.¹³

$$\kappa_e = \exp \left[2222 \left(\frac{1}{303} - \frac{1}{T} \right) \right] (0.5139\lambda - 0.326) \quad [13]$$

Per Eq. 13, both the hydration of the membrane after water uptake during the cold start operation and the rising cell temperature improve the proton conductivity of the membrane and hence reduce the ohmic loss. Consequently, it is seen in Fig. 2c that the cell voltage first increases and then decreases during current-ramping start-ups with the ramping angle β being smaller than 50°. The decrease in cell voltage results from the intraelectrode ice blocking the pathways for oxygen transport as well as covering the electrochemical active area, leading to larger activation and mass-transport voltage losses. In contrast, a different trend is observed for the current-ramping start-ups with larger ramping rates ($\beta = 60^\circ$ – 80°). The cell voltage is seen to decrease first, even though the initial dry membrane is hydrated by the product water from ORR and the cell is warming up. This is caused by the overly rapid increase in current density. Although the membrane hydration and the rising cell temperature yield a smaller ohmic resistance, the large current density still results in a higher ohmic loss. After some time into the start-up process, the effects of rapid temperature rise and the further hydrated membrane take over the increase in current density, resulting in an increasing cell voltage. The cell voltages in all the three cases of $\beta = 60^\circ$ – 80° eventually drop rapidly as the cold start processes fail to succeed.

$I_0 = 50 \text{ mA/cm}^2$.— The case studies for a smaller initial current density of 50 mA/cm² (which is half of that in the baseline case) were also performed, and the results are presented in Fig. 3. Of the nine cases studied (with $\beta = 0, 10, 20, 30, 40, 50, 60, 70, 80$), none suc-

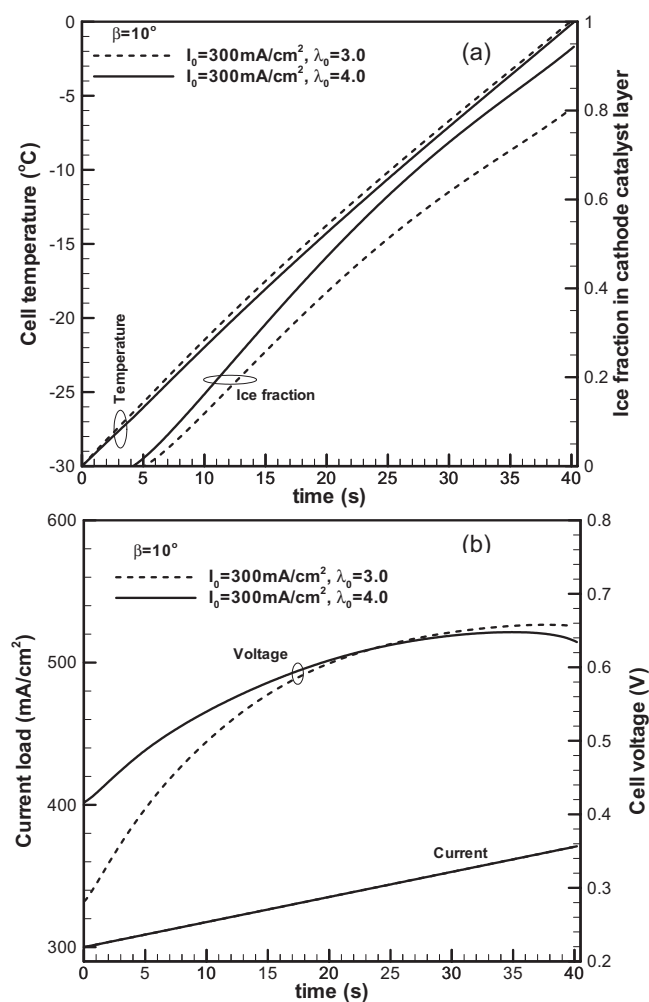


Figure 5. Current-ramping start-ups with initial current density of 300 mA/cm². (a) Cell temperature and ice fraction in CCL vs time and (b) current density and cell voltage vs time.

ceeds because the restricted initial current density cannot provide sufficient heat to warm up the cell to 0°C before ice completely fills the pores of the CCL.

$I_0 = 200 \text{ mA/cm}^2$.—A set of case studies with a larger initial current density of 200 mA/cm² (which is twice of that in the baseline case) was also conducted, and results are shown in Fig. 4 for a full range of ramping angles β varying within 0–80°. Clearly in Fig. 4, the PEMFC can be successfully started as long as the ramping angle is not greater than 40°. The larger initial current density enhances heat generation right from the beginning of start-up. Consequently, even if there is no current density ramping ($\beta = 0^\circ$), the cell can succeed in a cold start. The quickest start-up in this series takes ~40 s, which corresponds to a current-ramping angle of 40°.

A large initial current density (200 mA/cm²) combined with a fast ramping rate ($\beta = 80^\circ$) leads to a sharp drop in cell voltage (see Fig. 4c) upon loading as the MEA is initially dry (i.e., $\lambda_0 = 3$). This may cause a severe degradation of MEA and must be avoided in practice.

$I_0 = 300 \text{ mA/cm}^2$.—From the results presented in Fig. 2 and 4, it can be concluded that increasing the initial current density can shorten the start-up time. However, an excessive initial current density can lead to overly low cell voltage upon loading, which is detrimental to the MEA. In practice, an optimum start-up should not only have a short start-up time, but it should also avoid operating in

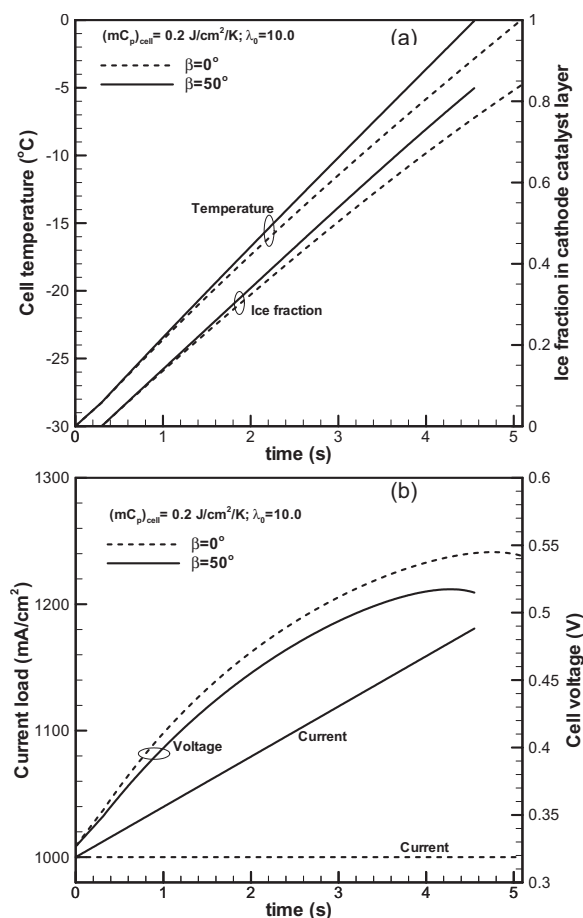


Figure 6. Current-ramping start-ups of a PEMFC with reduced thermal mass, 0.2 J/cm² K. (a) Cell temperature and ice fraction in CCL vs time and (b) current density and cell voltage vs time.

a very low voltage, which can cause MEA degradation. Evident from Eq. 13, the ohmic loss due to membrane resistance can be minimized, and a reasonably high cell voltage can be achieved provided that the membrane water content is sufficiently high initially.

Using an initial current density of 300 mA/cm², we performed two case studies with the initial membrane water content λ_0 being 3.0 and 4.0. Results are shown in Fig. 5. The PEMFC can be successfully started up in ~40 s. The start-up with an initially wetter membrane has an initial cell voltage higher than 0.4 V, which is more acceptable from the viewpoint of preventing MEA degradation.

$(mC_p)_{\text{cell}} = 0.2 \text{ J/cm}^2 \text{ K}$.—Finally, a rapid start-up of a next-generation PEMFC with a reduced thermal mass of 0.2 J/cm² K was attempted. Due to the thermal mass being halved, the cell temperature can rise faster when using the linear current-ramping start-up strategy with a relatively large initial current density. Consequently, the water absorption from the CCL into the membrane would be much faster than that with the standard thermal mass, rendering a small likelihood for IIF.

Using a very large initial current density of 1 A/cm², starting up the PEMFC with a reduced thermal mass of 0.2 J/cm² K was explored. To get the cell operational at the beginning of cold start and to avoid an overly low cell voltage, the MEA was taken to be relatively wet or hydrated with $\lambda_0 = 10.0$ before the cold start. The results presented in Fig. 6 indicate that the PEMFC can be successfully started up in about 5 s. A comparison between two cases with

different ramping rates indicates that for an ultrarapid start-up, current density ramping is unnecessary as the whole start-up process occurs within such a short time period.

The present model is for a generic cell in a stack with a varying heat loss coefficient. One may use this cell model with different heat loss coefficients to model both end cells (e.g., large heat loss) and middle cells (e.g., adiabatic condition) and to subsequently formulate a start-up strategy to apply to an overall stack. Thus, the cell model presented here could be incorporated into a stack model.

Conclusions

A current-ramping strategy has been proposed for rapidly starting up PEMFCs from a subfreezing environment. This strategy is made effective by taking advantage of the rising cell temperature and improved water adsorption of the membrane during cold start, thereby permitting an aggressive increase in the operating current density with time without forming much ice in the CCL. The ramping current density, in turn, generates more heat and accelerates the cell warm-up. Within the framework of a linear current-ramping start-up, we have explored the limits of the shortest start-up time and optimized materials properties and operating conditions needed to achieve rapid cold starts.

Generally, a larger initial current density and a higher current-ramping rate accelerate heat generation and facilitate rapid starts. However, an overly large initial current density and an overly high current-ramping rate also promote ice formation in the CCL, thus leading to a shutdown. For a PEMFC with standard cell thermal mass ($0.4 \text{ J/cm}^2 \text{ K}$) and sufficiently dry MEA initially ($\lambda_0 = 3.0$), either an initial current density of 100 mA/cm^2 coupled with an intermediate ramping rate ($\beta = 30\text{--}50^\circ$) or a larger initial current density of 200 mA/cm^2 in combination with a small ramping rate ($\beta \leq 40^\circ$) can lead to successful self-starts from the -30°C environment. However, a current-ramping cold start with an overly small initial current density (e.g., 50 mA/cm^2) shuts down with any current-ramping rate. Keeping the MEA mildly hydrated before a cold start can be favorable to rapid start-up strategies, as this makes the application of a large initial current density practical without an excessive voltage dropdown. Finally, a next-generation PEMFC with a reduced thermal mass of $0.2 \text{ J/cm}^2 \text{ K}$ and a relatively hy-

drated MEA can be successfully started up from -30°C in about 5 s provided that it is feasible to draw an initial current density as high as 1 A/cm^2 .

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References

1. Y. Hishinuma, T. Chikashisa, F. Kagami, and T. Ogawa, *JSME Int. J., Ser. B*, **47**, 235 (2004).
2. M. Oszcipok, D. Riemann, U. Kronenwett, M. Kreideweis, and M. Zedda, *J. Power Sources*, **145**, 407 (2005).
3. L. Mao, K. Tajiri, S. Ge, X. G. Yang, and C. Y. Wang, Abstract 998, The Electrochemical Society Meeting Abstracts, Vol. 502, Los Angeles, CA, Oct 16–21, 2005.
4. S. Ge and C. Y. Wang, *Electrochem. Solid-State Lett.*, **9**, A499 (2006).
5. L. Mao and C. Y. Wang, *J. Electrochem. Soc.*, **154**, B139 (2007).
6. K. Tajiri, Y. Tabuchi, and C. Y. Wang, *J. Electrochem. Soc.*, **154**, B147 (2007).
7. L. Mao, C. Y. Wang, and Y. Tabuchi, *J. Electrochem. Soc.*, **154**, B341 (2007).
8. K. Tajiri, Y. Tabuchi, F. Kagami, S. Takahashi, K. Yoshizawa, and C. Y. Wang, *J. Power Sources*, **165**, 279 (2007).
9. S. Ge and C. Y. Wang, *Electrochim. Acta*, **52**, 4825 (2007).
10. J. Hou, H. Yu, B. Yi, Y. Xiao, H. Wang, S. Sun, and P. Ming, *Electrochem. Solid-State Lett.*, **10**, B11 (2007).
11. E. L. Thompson, J. Jorne, and H. A. Gasteiger, *J. Electrochem. Soc.*, **154**, B783 (2007).
12. S. Ge and C. Y. Wang, *J. Electrochem. Soc.*, **154**, B1399 (2007).
13. F. Jiang, W. Fang, and C. Y. Wang, *Electrochim. Acta*, **53**, 610 (2007).
14. J. Li, S. Lee, and J. Roberts, *Electrochim. Acta*, **53**, 5391 (2008).
15. E. L. Thompson, J. Jorne, W. B. Gu, and H. A. Gasteiger, *J. Electrochem. Soc.*, **155**, B625 (2008).
16. E. L. Thompson, J. Jorne, W. B. Gu, and H. A. Gasteiger, *J. Electrochem. Soc.*, **155**, B887 (2008).
17. F. Jiang and C. Y. Wang, *J. Electrochem. Soc.*, **155**, B743 (2008).
18. C. Y. Wang, X. G. Yang, Y. Tabuchi, and F. Kagami, in *Handbook of Fuel Cells*, Vol. 6, W. Vielstich, H. Yokokawa, and H. A. Gasteiger, Editors, John Wiley & Sons, New York (2009).
19. U.S. Department of Energy, Progress Report for Hydrogen, Fuel Cells, and Infrastructure Technologies Program, Appendix A (2002).
20. K. G. Gallagher, B. S. Pivovar, and T. F. Fuller, *J. Electrochem. Soc.*, **156**, B330 (2009).
21. T. E. Springer, T. A. Zawodinski, and S. Gottesfeld, *J. Electrochem. Soc.*, **138**, 2334 (1991).